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1. Diabetes Progression Estimation (Regression)

Aim: Predict the progression of diabetes (a continuous disease progression measure) from diagnostic variables.

Dataset:

- **Diabetes Dataset (Scikit-learn / Kaggle)**
- Kaggle version (easier to download as CSV):
  Diabetes Dataset (Kaggle)

Core Task:

- Use regression models to estimate a continuous diabetes progression score from biomedical predictors.

Suggested Models:

- Baseline: Linear Regression, Ridge, Lasso
- Nonlinear: Random Forest Regressor, Gradient Boosting Regressor

Key Deliverables:

1. **EDA & Data Cleaning Report**
   - Description of each feature, distributions, outliers.
   - Correlation heatmap with target variable.
2. **Modeling Notebook**
   - Training/validation split, cross-validation.
   - Hyperparameter tuning for at least one model.
3. **Performance Evaluation**
   - **Metrics:**

MAE ,
RMSE,
R2R2

- Comparison table of all models.

4. **Short Technical Report (APA Style)**
   - Problem definition, methodology, results, and limitations.
5. **Reproducible Code Repository**
   - Clear `README` with how to run experiments.

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2. Bike Sharing Demand Prediction (Time-Series / Regression)

Aim: Predict the number of rental bikes required at a given time based on weather and temporal features.

Dataset:

- **Bike Sharing Dataset (UCI via Kaggle mirror)**
- Bike Sharing Demand (Kaggle)

Core Task:

- Build models to estimate bike rental counts for given date-time and weather conditions.

Suggested Models:

- Linear Regression / Ridge
- Random Forest Regressor
- Gradient Boosting (XGBoost, LightGBM, or scikit-learn's GradientBoostingRegressor)

Key Deliverables:

1. **Time-Series EDA**
    - Plots of demand vs. hour/day/season.
    - Discussion of seasonality and trends.
2. **Feature Engineering Report**
    - Extract hour, day of week, holiday, working day, etc.
    - Encode categorical variables.
3. **Model Training & Comparison Notebook**
    - Cross-validation.
    - Feature importance analysis.
4. **Error Analysis**
    - Residual plots over time and across hours of day.
    - Identification of under/over-predicted periods.
5. **Interactive Visualization (Optional but Good)**
    - Simple dashboard or set of plots showing predicted vs. actual demand.
6. **Final Report & Presentation**
    - Focus on how the model could support operational decision-making.

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3. Air Quality Index (AQI) / PM2.5 Estimation (Regression)

Aim: Estimate air pollution level (e.g., PM2.5 concentration) using meteorological and other variables.

Dataset:

- **Beijing PM2.5 Data (UCI via Kaggle)**
  Beijing PM2.5 Data (Kaggle)

Core Task:

- Predict PM2.5 values (or AQI) for a monitoring station using weather and time features.

Suggested Models:

- Linear Regression, Random Forest Regressor
- Gradient Boosting Regressor
- (Advanced/optional) LSTM for sequence modeling

Key Deliverables:

1. **Data Understanding & Cleaning**
   - Handling missing data, time indexing.
   - Visualizing temporal patterns and extremes.
2. **Baseline vs. Advanced Models**
   - Baseline: Predict mean, last value, or persistence model.
   - ML models: regressors with feature engineering (lags, rolling means).
3. **Model Evaluation & Comparison**
   - Metrics: MAE, RMSE,

R2

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- Seasonal performance analysis (winter vs summer).

4. **Feature Importance & Interpretation**
   - Which variables most strongly affect PM2.5?
   - Partial dependence / simple interpretation.
5. **Policy-Relevance Summary**
   - 1–2 pages: how such a model could help cities (alert systems, planning).
6. **Code + Notebook**
   - Clean, commented notebook(s) for full pipeline.

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4. Real Estate Price Estimation (Regression)

Aim: Predict house prices from structural and location-based features.

Dataset Options (you can choose one):

1. **Ames Housing Dataset (Kaggle)** – richer than the classic Boston dataset
   Ames Housing Dataset (Kaggle)
2. **House Prices – Advanced Regression Techniques (Kaggle)**
   House Prices (Kaggle)

Core Task:

- Build a regression model to estimate house sale prices.

Suggested Models:

- Linear Regression, Ridge, Lasso
- Random Forest Regressor
- Gradient Boosting / XGBoost

Key Deliverables:

1. **EDA & Feature Engineering**
   - Handling categorical features (encoding).
   - Dealing with missing values, skewed features (log transform).
2. **Model Benchmarking**
   - Train at least 3–4 different regressors.
   - Use k-fold cross-validation.
3. **Residual & Error Analysis**
   - Error distribution analysis (overpricing vs underpricing).
   - Plots of predicted vs actual prices.
4. **Deployed Prediction Service (Basic)**
   - Simple API (Flask/FastAPI) or a notebook UI that takes features and returns price.
5. **Technical Report**
   - Discuss generalization, overfitting, feature selection impacts.
6. **Reproducibility**
   - Requirements file, clear `README`, versioned code.

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5. Customer Churn Prediction (Classification)

Aim: Predict whether a customer will churn (leave the service) based on usage and account information.

Dataset Options (choose one):

1. **Telco Customer Churn (IBM / Kaggle)**
   Telco Customer Churn (Kaggle)
2. **Bank Customer Churn Modeling (Kaggle)**
   Bank Churn Model (Kaggle)

Core Task:

- Binary classification: churn vs not churn.

Suggested Models:

- Logistic Regression
- Random Forest / Gradient Boosting (XGBoost)
- (Optional) Simple Neural Network

Key Deliverables:

1. **EDA & Class Imbalance Analysis**
   - Churn rate, distribution of features by class.
   - If imbalance exists, apply techniques (SMOTE, class weights).
2. **Model Training & Tuning**
   - Train at least 3 models.
   - Hyperparameter search for the best model.
3. **Evaluation & Business Metrics**
   - Metrics: Accuracy, Precision, Recall, F1, ROC-AUC.
   - Confusion matrix and discussion at a chosen probability threshold.
4. **"Retention Strategy" Memo**
   - Interpretation of important features.
   - How the business can act on high-churn-risk customers.
5. **Scoring Pipeline**
   - Function or small script that takes new customer data and outputs churn probability.
6. **Final Report & Slides**
   - Non-technical explanation for a manager-level audience.

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6. Credit Default Risk Prediction (Classification)

Aim: Predict whether a borrower will default on a credit card or loan based on demographic and behavioral features.

Dataset:

- **Default of Credit Card Clients Data Set (UCI via Kaggle)**
  Default of Credit Card Clients (Kaggle)

Core Task:

- Binary classification: default vs non-default.

Suggested Models:

- Logistic Regression (baseline)
- Random Forest / Gradient Boosting (XGBoost/LightGBM)
- (Optional) Simple Neural Network

Key Deliverables:

1. **Data Preprocessing Pipeline**
   - Handling categorical variables, scaling numeric features.
   - Train/validation/test split.
2. **Model Comparison & Calibration**
   - ROC curves for multiple models.
   - Calibration plots or Brier score if possible.
3. **Risk Scoring Framework**
   - Convert model output probabilities into risk bands (low/medium/high).
4. **Fairness & Ethics Section**
   - Brief discussion of potential bias, fairness concerns, and responsible use.
5. **Model Card / One-Page Summary**
   - Very concise documentation of data, model, metrics, intended use, limitations.
6. **Code & Documentation**
   - Clear scripts/notebooks and instructions.

Common Course-Level Deliverables (For All Projects)

You can standardize these across the course:

1. **Project Proposal (2–3 pages)**
    - Problem definition, dataset description (with URL), planned methods, expected outcomes.
2. **Midterm Progress Report**
    - Cleaned dataset, preliminary EDA, and baseline model results.
3. **Final Technical Report (APA Style)**
    - Abstract, introduction, related work, methodology, experiments, results, conclusion, references.
4. **Oral Presentation (10–15 minutes)**
    - Problem, approach, main findings, and limitations.
5. **Code Repository**
    - Version-controlled (Git), with `README`, clear folder structure, and environment requirements.
6. Prototype 1
7. Prototype 2

Discovery, Development, Refinement, and Presentation. This structure prevents "last-minute" coding and encourages the scientific method.

13-Week Project Roadmap

Phase	Week	Milestone	Key Activities & Deliverables
I: Initiation	1-2	**Project Selection & Proposal**	Form groups, select one of the 6 subjects, and explore the dataset. **Deliverable:**2-page Proposal (Problem statement, Dataset link, and Team roles).
	3	**Literature & Context**	Research similar ML applications. **Deliverable:** Annotated bibliography (3-5 sources in APA style).
II: Data	4-5	**Exploratory Data Analysis (EDA)**	Data cleaning, handling missing values, and visualization. **Deliverable:** EDA Report (Plots of distributions, correlations, and outliers).
	6	**Feature Engineering**	Scaling, encoding, and creating new features (e.g., lag variables for Bike Sharing). **Deliverable:** Preprocessing Pipeline script.
III: Modeling	7	**Baseline Modeling**	Implement simple models (Linear/Logistic Regression). **Deliverable:** Baseline Performance Table (MAE/RMSE or Accuracy/F1).
	8-9	**Advanced Modeling**	Implement Random Forest, XGBoost, or LSTMs. Perform Hyperparameter tuning (GridSearch/RandomSearch).
	10	**Model Evaluation**	Cross-validation, Residual analysis, and Confusion Matrices. **Deliverable:**Midterm Progress Presentation (Internal).
IV: Finalize	11	**Interpretation & Bias**	Analyze feature importance (SHAP/LIME) and check for model bias.
	12	**Documentation & Packaging**	Finalize the code repository and write the technical report. **Deliverable:** GitHub Repository with `requirements.txt` and `README`.
	13	**Final Delivery**	**Deliverables:** Final Technical Report (APA) and Final Presentation/Demo.

Grading Rubric (Total: 100 Points)

Category	Weight	Criteria
Data Quality	20%	Depth of EDA, appropriate handling of missing data/outliers, and creative feature engineering.
Technical Execution	30%	Correct implementation of at least 3 models, proper validation (Train/Test/CV), and hyperparameter tuning.

Category	Weight	Criteria
Evaluation & Analysis	20%	Use of correct metrics (

R2R2

, RMSE, F1, AUC), error analysis, and interpretation of results. |
| **Documentation** | 15% | APA-style report, code readability, comments, and a professional GitHub `README`. |
| **Presentation** | 15% | Clarity of communication, quality of slides, and ability to answer technical questions. |

Subject-Specific Deliverable Checklist

For the 6 subjects previously defined, ensure students include these specific items in their Week 12/13 package:

1. **Diabetes Progression:** A plot showing "Predicted vs. Actual" progression with a 45-degree reference line.
2. **Bike Sharing:** A visualization of demand "Heatmap" by hour of day and day of week.
3. **Air Quality (PM2.5):** A "Time-Series Residual Plot" to show if the model fails during specific seasons.
4. **Real Estate:** A "Feature Importance Bar Chart" showing the top 10 factors affecting house prices.
5. **Customer Churn:** A "Cost-Benefit Analysis" (e.g., "If we give a 10% discount to those predicted to churn, how much revenue is saved?").
6. **Credit Default:** A "Precision-Recall Curve" to discuss the trade-off between missing a defaulter vs. annoying a good customer.

References for Course Design:

- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning*. Springer.
- Zhang, A., Lipton, Z. C., Li, M., & Smola, A. J. (2023). *Dive into Deep Learning*. Cambridge University Press.